Network Management

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1.

Performance management is important in a cloud platform because many users and applications depend on the system working quickly and smoothly. A performance management setup monitors how well the system is working and helps administrators detect problems early. In a cloud platform, monitoring tools are used to check things such as CPU usage, memory usage, network traffic, and storage performance.

The system collects data from servers and network devices continuously. This information is analyzed to identify slow responses, heavy traffic, or system overload. If performance drops below a certain level, alerts are sent to administrators so they can fix the issue quickly.

For example, if many users are accessing the platform at the same time, the system can automatically allocate more resources such as processing power or memory. This helps keep the system stable and fast.

Performance reports are also created regularly to help administrators understand long-term trends. These reports help improve planning and ensure the cloud platform continues to run efficiently and reliably.

2.

DiffServ stands for Differentiated Services. It is a network management method used to control and improve the quality of network traffic. DiffServ helps manage how different types of data move through a network by giving them different priority levels.

In a network, not all data needs the same level of speed or reliability. For example, voice calls and video meetings require fast and stable connections, while emails or file downloads can tolerate slight delays. DiffServ allows network administrators to classify and mark packets of data so routers know how to treat them.

Each packet is marked with a specific code in the IP header called the DSCP (Differentiated Services Code Point). Routers then use this code to decide how quickly the packet should be forwarded.

This system improves network performance because important data is prioritized. As a result, real-time services like video calls, online gaming, and streaming can work more smoothly even when the network is busy.

3.

The tc (traffic control) tool in Linux is used to manage and control network traffic. It allows administrators to limit bandwidth, prioritize certain types of traffic, and simulate network conditions such as delay or packet loss.

To use tc, administrators configure rules that control how packets move through the network interface. For example, tc can be used to limit download speeds for certain applications or give higher priority to important services like video conferencing.

Using tc helps improve network performance and fairness between users. If one user or application uses too much bandwidth, it can slow down the network for everyone else. With tc, administrators can prevent this problem by controlling traffic usage.

Another advantage of tc is that it helps test network behavior. Developers can simulate slow networks or delays to see how applications perform. This helps improve application reliability and user experience in real-world conditions.

4.

A performance management policy is a set of rules used to maintain good performance in an IT system. The purpose of this policy is to ensure that systems run efficiently and that resources are used properly.

The policy usually defines performance standards such as response time, network speed, and system availability. It also explains how the system should be monitored and what actions should be taken when performance problems occur.

For example, the policy may state that CPU usage should not exceed a certain percentage for long periods. If the limit is reached, administrators must investigate and solve the issue.

The policy may also include regular monitoring, performance reporting, and automatic alerts. These measures help detect issues early before they affect users.

Overall, a performance management policy helps maintain stable systems, improve user experience, and support better planning for future system growth.